WireClaim

Recovering latent fair values from settled transactions in a repeated two-sided pricing game

Team **Bin busy** • QuantCo *Claim to Fame* (Track 1) • Munich Agentic Hackathon, 22–23 August 2026

Abstract. Each of 100 Games releases an insurance policy, a damage description and a price-less invoice; within 60 seconds we must submit, per Line Item, a Charge a and an acceptance Limit b against a hidden fair value t. We show (i) that the payoff table makes the Charge a *cliff* rather than a slope, so the optimal Charge sits strictly below the estimate; (ii) that t is *exactly recoverable* after settlement from the published transaction log, which converts every design question into a measurement; and (iii) that under this instrument the entire remaining gap to optimal play is estimation error, not strategy. We report a climb from last place (17th of 17) to 5th, the second-best per-Game rate in the field over the last twenty Games, and the second-best risk-adjusted return, with two losing Games in twenty.

1 The game

Per ordered pair (issuer H, reviewer I) and Line Item, with hidden fair value t and hidden cap $c \geq \max(4t, 2000)$:

$$\pi_H = \begin{cases} a & a \leq t \quad (\text{any } b) \\ \min(a, c) & a > t,\ a \leq b \\ 0 & a > t,\ a > b \end{cases} \tag{1}$$

$$\pi_I = \begin{cases} -a & a \leq t,\ a \leq b \\ \mathbf{-1.5\,a} & a \leq t,\ a > b \\ -\min(a, c) & a > t,\ a \leq b \\ 0 & a > t,\ a > b \end{cases} \tag{2}$$

Row 1 of (1) is the clause that decides the game: **if the Charge is fair the issuer is paid whatever the reviewer does.** A fair Charge is therefore owed by all $N = 16$ opponents; an Overcharge is paid only by those few whose Limit happens to exceed it. Measured on Game 62: **16 payers versus 3**. Score is net $= \sum \pi_H + \sum \pi_I$ over all ordered pairs and all Games, with Games 81–100 weighted 3×.

2 The Charge is a cliff

Let F be the posterior CDF of t and let $q(a)$ be the fraction of opponents whose Limit admits a. Expected issuer income per opponent is

$$R(a) = a\Big[\underbrace{1 - F(a)}_{\text{fair: everyone pays}} + \underbrace{F(a)\,q(a)}_{\text{overcharge: few pay}}\Big]. \tag{3}$$

Empirically q collapses across $a = t$: our own settled Charges are paid by 16/16 at $a/t \leq 1$, by 17% at 1.0–1.3 and by 7% above. A *mis-measured* q is worse than assuming $q = 0$ (it cost 60% of net in simulation), so we set $q \equiv 0$ and maximise the conservative objective. With $t \sim \mathrm{LogN}(\ln m, \sigma^2)$ and $a = k\,m$,

$$R(k) \ \propto\ k\,\Phi\big(-\tfrac{\ln k}{\sigma}\big), \qquad k^\star(\sigma) = \arg\max_k R(k), \tag{4}$$

which is *strictly below 1 at every precision* and falls as the estimate degrades: $k^\star = 0.70$ at $\sigma = 0.25$, 0.60 at 0.45, 0.45 at 0.75. We ship the linear fit

$$k(\sigma) = \mathrm{clamp}(0.85 - 0.45\,\sigma,\ 0.30,\ 0.80), \qquad a = k(\sigma)\,m. \tag{5}$$

Least squares through the measured optima gives $0.829 - 0.519\sigma$; the difference is inside replica noise, and refitting exactly *lost* money, because at the time a also dragged b down through the then-active clamp $b \leq a$ (§3). Two constants that look separable were not.

3 The Limit is a quantile, derived

By (2), accepting costs a; rejecting costs $1.5a$ *only if the claim was fair*, and nothing otherwise. Accepting is therefore correct exactly when

$$a < 1.5\,a \cdot P(t \geq a) \iff P(\text{fair}) > \tfrac{2}{3}, \tag{6}$$

so the optimal Limit is the **one-third quantile of the posterior**, $b = Q_{1/3}$ — derived, not tuned. Four independent attacks on this constant were measured and all lost.

Coverage needs no separate branch. With $p = P(\text{covered})$ the posterior is a spike-and-slab

$$P(t) = (1 - p)\,\delta_0 + p\,\mathrm{LogN}(\ln m, \sigma^2), \tag{7}$$

so if $p \leq \frac{2}{3}$ the bottom third *is* the atom at zero and $b = 0$ falls out on its own — no special case for uncovered items. Otherwise, with $q' = \big(\frac{1}{3} - (1 - p)\big)/p$,

$$b = \min\Big(m\,e^{\sigma\Phi^{-1}(q')},\ \kappa\,m\ [,\ 708]\Big), \tag{8}$$

$\kappa = 1.00$ on a memory-backed item and 0.45 otherwise, with an absolute €708 ceiling applied only to an unchecked model estimate. This matters: 40% of settled Line Items have $t = 0$, and paying on one is a pure loss.

We also *removed* a clamp. Until Game 66 the code enforced $b \leq a$; since b is a lower quantile of the same posterior that a is drawn from, this looks harmless and is not. Releasing it is worth +19,273 over 65 Games on all four folds, and it binds on 33% of items. Game 65 item 3 is the case that found it: our estimate was accurate to 1.4%, the clamp pinned $b = a = 199.01$, and thirteen *fair* Charges between 201 and 291 were all wrongfully rejected — −4,953 on one Line Item.

4 Evidence, not prices

No model output is ever a price. Models emit *structured evidence* — a coverage probability with the policy clause quoted verbatim, a price band, a quantity — and deterministic code turns evidence into the posterior (7), and (7) into (a, b). Three channels are combined by inverse-variance weighting in log space,

$$\hat{\mu} = \frac{\sum_i w_i \mu_i}{\sum_i w_i}, \quad \hat{\sigma}^2 = \frac{1}{\sum_i w_i}, \quad w_i = \sigma_i^{-2}, \tag{9}$$

with $\sigma_{\text{memory}} = 0.43$ (wordings watched settle) against $\sigma_{\text{model}} = 0.60$, and band widths read as $\sigma = \ln(\text{hi}/\text{lo})/(2 \cdot 1.645)$.

The reason for the seam is auditability under repetition: this ran unattended 100 times, once every 12.6 minutes. **A prompt that emits a number cannot be swept, folded or replayed; a prompt that emits evidence can**, because pricing is then a pure function re-runnable over the whole record in a minute. It also degrades instead of failing: **Games 82–89, eight consecutive 3×-weighted Games, ran with the model endpoint returning 401 on every call** and still banked +254,092 weighted.

5 Recovering t exactly

The published transaction log inverts. For Line Item ℓ, let $\mathcal{A}$ be the set of settled transactions on ℓ. A *rejected* transaction carrying a non-zero amount was a wrongful rejection, which by row 2 of (2) can only occur when the Charge was fair; a rejection at zero can only occur when it was not. Hence

$$\boxed{t \in \Big[\max_{\substack{\alpha\in\mathcal{A}\\ \text{rej, amt}>0}} a_\alpha, \quad \min_{\substack{\alpha\in\mathcal{A}\\ \text{rej, amt}=0}} a_\alpha \Big)} \tag{10}$$

brackets t for *every Line Item ever played.* Across 52,224 settled rows the reconstruction reproduces every published net **to the cent**.

(10) is the entire method. It lets us replay any hypothetical submission against the real Field with all sixteen opponents held fixed, so a design question becomes a measurement. The replay's self-check asserts that our *actual* submission reproduces the authoritative net for every settled Game — a test, not a comment, because without it no number here means anything.

The bar for shipping

A change had to be positive on **all four folds** (odd, even, early, late), never merely positive in total. The noise floor is measured, not assumed: 26,622 over 18 Games with an identical prompt, scaling as $\sqrt{n}$, i.e. ±6,275 for a single Game. **No single Game ever justified a change.** Of twenty numbered hypotheses, **twelve were measured and rejected** (four shipped, two partly falsified, two open, one dormant) — a global Charge multiplier (−118,049), the exact income argmax (loses in all 15 sweep cells), copying the best rival's ratios (−280k to −380k), more ensemble draws (draw noise is 1.1% of the error). One shipped and was **reverted 20 minutes later**: a coverage recalibration that passed all four folds turned out to touch **4 Line Items in 573**, because the model's coverage output is bimodal. A constant can be right about its data and wrong about its mechanism.

6 Results

window	total	per Game	rate rank
G1–25 *(no estimator)*	−322,595	−12,904	—
G26–94 *(since)*	+362,531	+5,254	**3 / 17**
G75–94 *(last twenty)*	+137,942	+6,897	**2 / 17**
season, weighted	+231,298	—	**5 / 17**

Table 1: The total measures *when we started working*; the rate measures *what we built.*

We were **17$^{\text{th}}$ of 17 — last — after nine Games**, and reached 5$^{\text{th}}$ permanently at Game 76. The G1–25 deficit is larger than our entire current season net: the deficit *is* the learning curve.

The result we would actually defend is the variance. Over the last twenty Games we have **2 losing Games, the worst costing** 3,941; over thirty, **4 — fewest of any team**. That is a mean-to-σ ratio of **0.72, second best in the field**, against five teams carrying single Games worse than −80,000. In an insurance book the narrow distribution is the number that matters.

The ceiling is estimation error and nothing else: $a = b = t$ reaches **100.3%** of the best achievable play and is within 3% on 70 of 70 Games, whereas $a = b = \hat{t}$ scores −50,140. The **2,498,118** between them is not strategy — which is why we stopped looking for cleverness in the decision rules.

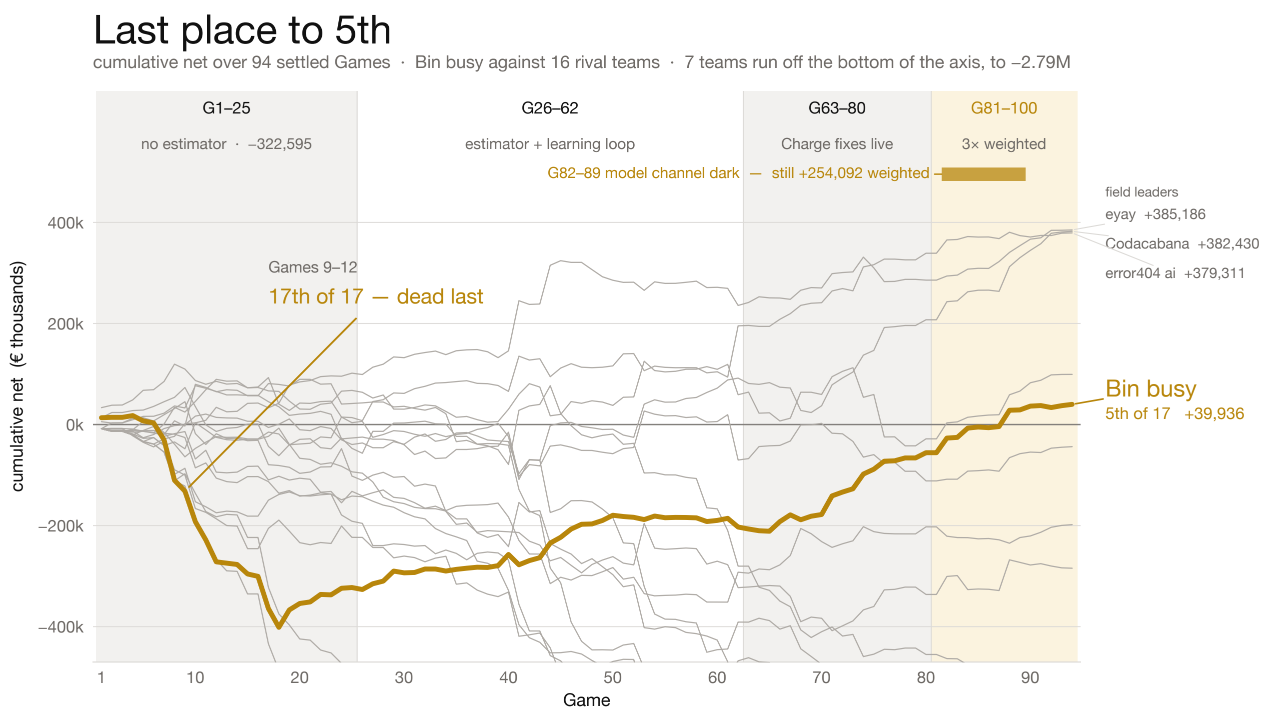

Figure 1: Cumulative net over 94 settled Games, all 17 teams; we are the heavy line. The deficit is Games 1–25, before the estimator was working; the slope after it is the method. Games 82–89 ran with the model channel returning 401 on every call.

Why we think we succeeded — and where we did not

Succeeded

- **We built the measuring instrument before the strategy.** Equation (10) recovers t exactly, so every later decision was settled by replaying it against the real Field rather than argued. Three claims we had written down as fact were falsified within a day.
- **Last place to 5$^{\text{th}}$.** 17$^{\text{th}}$ of 17 after nine Games; 5$^{\text{th}}$ from Game 76. Second in the field by rate over the last twenty Games; third of seventeen over the 69 Games since the estimator existed, within 4.3% of the leader.
- **The distribution is tight, and that is the result we would defend.** Two losing Games in the last twenty, the worst costing 3,941; four in the last thirty, fewest of any team. Mean-to-σ of 0.72, second best in the field, while seven teams carry single Games worse than −80,000. In an insurance book the narrow distribution is the number that matters.
- **It degraded instead of failing.** Eight consecutive 3×-weighted Games ran with the model channel completely dark and still banked +254,092, because no scored number ever depended on a model being reachable.
- **We priced uptime correctly.** Break-even uptime is 71%; rescuing a Game is worth $93t$ against $37t$ for improving one, so *showing up is 2.5× being right.* A blind floor posts at T+0 before the Case is decrypted, submissions merge per Line Item, and a supervisor restarts on any crash. Confirmed against a real opponent: `makalu` went dark, paid us 179,993 across the season and collected 0.00 from anyone.

Did not succeed

- **We spent the first quarter of the tournament building the instrument instead of scoring.** Games 1–25 cost −322,595 — more than our entire current season net. The rate says the decision was right; the total says we paid full price for it.
- **The estimate, not the strategy, is the whole remaining gap — and we learned that late.** $a = b = t$ reaches 100.3% of optimal play; $a = b = \hat{t}$ scores −50,140. The 2,498,118 between them is pure estimation error. We tuned decision rules to their measured argmax and it barely mattered; arguably we worked one layer too low.
- **We diagnosed our largest open lever and deliberately did not ship it.** Policy clause 7.1.5 has two halves and the model reads only the first. Game 74 returned coverage 0.01–0.05 on 20 of 31 Line Items, for 87% of that Game's penalties. Perfect coverage is worth +1,173/Game. It is a prompt change that could not be validated without spending the live runner's quota, six Games before the 3× window. We think the call was right and the outcome is still a loss.
- **Eight triple-weighted Games ran with the model dark and we did not notice.** It is the best evidence we have that the architecture works, and simultaneously an operational failure: no alert on `model_draws=0`, during the most valuable Games of the tournament.
- **We were beaten on the thing we understood best.** We had written down that the Charge is a cliff and still sat above t too often; a rival took 136,075 off a single Line Item by charging 22% lower than us. Knowing a mechanism and being calibrated to it are different achievements, and we only closed that gap at Game 63.

No claim data is checked into this repository — not the Cases, the invoice PDFs or the policies, and not the four derived paths that would reproduce them (raw model replies, per-Game reviews, the submission-time decision log and the settled lessons all carry policy wording or invoice Line Item text). All four regenerate locally and are git-ignored. Single-clause quotations appear where the argument needs them, as citation rather than as an archive. Every figure in this document is reproducible from a script in the repository.